Distance on the Coordinate Plane

Teacher's Edition

The Period

MA.8.GR.1.2 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

Must Find the distance between two points: on one line by subtracting (across an axis too), slanted by the right triangle whose legs are read from the coordinates; an irrational answer exact and, when asked, to the tenth.

Must not The distance formula (Clarification 2); radicands over 225; perimeters and maps with several legs (6.06).

Target I can find the distance between two points: subtract on a line, and draw the right triangle when the segment is slanted.

MTR **MTR.2.1** — Notes II–III: one distance as a slanted segment, as a right triangle's legs and as $a^2 + b^2 = c^2$ · **MTR.4.1** — boards 3 and 6 split; the sixty-second re-vote · **MTR.6.1** — Notes I and board 9: a distance across zero checked by counting the squares

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	11 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	4 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. Same y: subtract the x's; same x: subtract the y's — across zero the two pieces add.
2. Slanted: draw the right triangle; its legs come from the coordinates.
3. A distance is never negative, and there is no new formula: the triangle is the method.

Slide by Slide

1. **Distance on the Coordinate Plane.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is Notes I's rule before it is written down: same x, so subtract the y's — and 5 − (−3) is 8, across zero. *1 retrieves 6.04 — a leg in a story (yesterday). 2 retrieves 6.01–02 — the converse (last week). 3 retrieves 5.01 and 5.04 — a real distance from a map's scale (last unit). 4 retrieves MA.6.GR.1.2 — the distance between points with the same x-coordinate (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 7 ft 2. Yes — 64 + 225 = 289 = 17^2 3. 17 km 4. 8
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy the rule and the example.** Across zero, the two pieces add.
6. **Notes.** Slanted? Draw the triangle. Across, then up.
7. **Notes.** Across: the x's. Up or down: the y's. Then the theorem.
8. **Example 1 — On one line.** Ask what the two points share before anyone subtracts: the x, so the distance is in the y's.
9. **Example 1 — Worked.** Same x, so the y's: 7 − (−2) = 9 — 2 below the axis and 7 above it.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same y: 8 − (−3) = 11.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — Slanted.** Ask whether the points share an x or a y: neither — so draw the right triangle first.
13. **Example 2 — Worked.** Across 4 − (−2) = 6, up 9 − 1 = 8; 36 + 64 = 100, so 10 units — whole, because 6, 8, 10 is a triple.
14. **Your Turn — Same steps, your numbers. Boards up when done.** Across 2 − (−5) = 7, up 1 − (−3) = 4; 49 + 16 = 65, √65 = 8.06, so 8.1.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).

16–17. Board 1. How far apart are $(-7, -4)$ and $(-7, 6)$? → ***10 units*** *2 — subtracted 6 − 4 across zero [Source of Truth, GR.1.2: integer errors across an axis]*

18–19. Board 2. How far apart are $(-9, 2)$ and $(3, 2)$? → ***12 units*** *6 — subtracted 9 − 3 across zero [Source of Truth, GR.1.2: integer errors across an axis]*

20–21. Board 3. What is the distance between (−5, 1) and (3, 1)? → ***B — 8 units***
(A) counted the grid lines, not the spaces [Math Nation names counting gridlines]; (C) subtracted 5 − 3 across zero [Source of Truth, GR.1.2: (−2, 0) to (3, 0) is not 1]; (D) −5 − 3: a distance is never negative [Notes I] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

22–23. Board 4. How far apart are $A(-4, -3)$ and $B(5, 9)$? → ***15 units*** *21 — added the legs [MA.8.GR.1.2: the theorem, not a sum]; 'cannot, no triangle' — draw one [Math Nation names 'no triangle drawn']*

24–25. Board 5. How far apart are $C(-2, -5)$ and $D(3, 3)$, to the nearest tenth? → ***√89 ≈ 9.4 units*** *13 — added the legs [MA.8.GR.1.2]; √(1² + 8²) — subtracted 3 − 2 across zero [Source of Truth, GR.1.2]*

26–27. Board 6. Which gives the distance between (4, 5) and (−3, −1)? → ***C — √(7² + 6²) = √85*** *(A) 4 − 3 = 1 for the across leg — the sign of −3 dropped [Source of Truth, GR.1.2: integer errors across an axis]; (B) one point's coordinates taken as the legs [Math Nation names wrong horizontal and vertical lengths]; (D) added the legs [MA.8.GR.1.2: the theorem, not a sum] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. How far is $(5, -12)$ from the origin? → ***13 units*** *17 — added the legs [MA.8.GR.1.2]; 'the square root of −119' — subtracted 25 − 144 [the theorem adds the legs' squares]*

30–31. Board 8. On a town map in miles, the school is at $(-3, 2)$, the library at $(4, 2)$ and the town hall at $(0, 5)$. How far is the school from the library? → ***7 mi*** *The the town hall at (0, 5) is not needed. 1 — subtracted 4 − 3 across zero [Source of Truth, GR.1.2]; a distance to the town hall — the wrong pair of points*

32–33. Board 9. Ty says (−3, 0) and (7, 0) are 4 units apart. What did he do? What is the distance? → ***He subtracted 7 − 3. Across zero the pieces add: 10 units.*** *'4 is right' — the axis-crossing error [Source of Truth, GR.1.2]*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Boards 1–2: one line, vertical then horizontal, both across zero; 3 votes on the axis error. The pattern breaks at 4 — slanted, with only the points on the grid — whole, then to the tenth (5), then only the expression (6), then from the origin (7); 8 carries an extra point; 9 is written. The independent set: horizontal, vertical, slanted whole, slanted to the tenth, from the origin, a map.

Misconceptions to Watch

Board 1. 2 — subtracted 6 − 4 across zero [Source of Truth, GR.1.2: integer errors across an axis]

Board 2. 6 — subtracted 9 − 3 across zero [Source of Truth, GR.1.2: integer errors across an axis]

Board 3. counted the grid lines, not the spaces [Math Nation names counting gridlines]

Board 4. 21 — added the legs [MA.8.GR.1.2: the theorem, not a sum]

Board 5. 13 — added the legs [MA.8.GR.1.2]

Board 6. 4 − 3 = 1 for the across leg — the sign of −3 dropped [Source of Truth, GR.1.2: integer errors across an axis]

Board 7. 17 — added the legs [MA.8.GR.1.2]

Board 8. 1 — subtracted 4 − 3 across zero [Source of Truth, GR.1.2]

Board 9. '4 is right' — the axis-crossing error [Source of Truth, GR.1.2]

Materials Whiteboards and markers; a printed grid for any student whose board has none. Calculators allowed, printed nowhere.